

A Study of Convolutional Neural Networks for Image Classification

Jane Doe, John Smith Department of Computer Science

Abstract

Deep learning has become the dominant approach for computer vision. In this paper we study convolutional neural networks and the attention mechanism for image classification. We show that batch normalization and dropout improve generalization and reduce overfitting on the test set.

1. Introduction

Machine learning models learn representations from data. A neural network is trained with stochastic gradient descent and backpropagation to minimize a loss function. Recent transformer architectures rely on self-attention to capture long-range dependencies between tokens in a sequence.

2. Method

Our encoder uses convolution and max pooling to extract a feature map. The learning rate and weight decay are tuned on a validation set. We apply data augmentation to the training set to improve generalization of the deep neural network.

The objective function is defined as:

$$L = - \sum_i y_i * \log(\text{softmax}(z_i)) + \lambda * || w ||^2$$

3. Results

Our model achieves state of the art accuracy. An ablation study confirms that the attention mechanism is important for the final precision and recall on the benchmark.

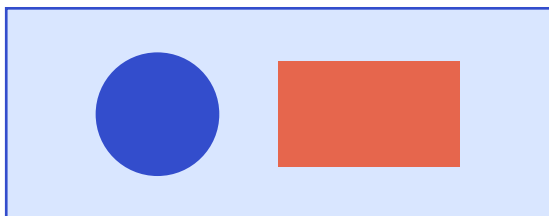


Figure 1: Architecture of the convolutional neural network.